

Force Formula Triangle

Use the cue, then fade the cue. Look at the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

CUE FOCUS The triangle links force (F), mass (m), and acceleration (a). Cover the unknown to read the formula: $F = m \times a$, $m = F \div a$, and $a = F \div m$.

Round 1 — Retrieve It

1. What three quantities does the force triangle connect?

2. Write the formula for force.

3. Write the formula for mass.

4. Which variable do you cover to find acceleration, and what formula results?

Round 2 — Sketch It

Draw the force triangle. Put force on top; put mass and acceleration on the bottom. Circle the two quantities you would multiply to find force.

Force — Apply It

Keep the cue card covered. Solve first. Then use the cue to check, revise, and explain what you changed.

Round 3 — Apply the Relationship

1. A 4 kg cart accelerates at 3 m/s^2 . What net force acts on it?

2. A force of 20 N gives an object an acceleration of 5 m/s^2 . What is its mass?

3. A net force of 18 N acts on a 6 kg block. What is its acceleration?

Round 4 — Reason from Evidence

The same net force of 24 N is applied to two carts. Cart A has a mass of 4 kg; Cart B has a mass of 8 kg. Use the triangle to find each cart's acceleration, then say which cart accelerates more and how you know.

Force — ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A

Articulate

Explain how the force triangle helps you find a missing quantity. Use one example that is not printed on the cue card.

C

Connect

Connect force to Newton's second law. Why does a larger net force on the same mass mean a larger acceleration, and how does that relate to $a = F \div m$?

E

Extend

Create your own problem that asks for acceleration. Give the numbers and show the work using $a = F \div m$.

Confidence check. Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it.

Teacher Key — Force

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. Force, mass, and acceleration.
2. $F = m \times a$.
3. $m = F \div a$.
4. Cover a ; $a = F \div m$.

Sketch It — look for

Force on top, with mass and acceleration on the bottom. Force is found by multiplying mass $\times$ acceleration (the two bottom quantities).

Apply It

1. $F = m \times a = 4 \times 3 = \mathbf{12\ N}$.
2. $m = F \div a = 20 \div 5 = \mathbf{4\ kg}$.
3. $a = F \div m = 18 \div 6 = \mathbf{3\ m/s^2}$.

Reason from Evidence — look for

Cart A: $a = 24 \div 4 = 6\ \text{m/s}^2$. Cart B: $a = 24 \div 8 = 3\ \text{m/s}^2$. Cart A accelerates more because the same net force acts on a smaller mass.

ACE — Extend look-for

Many answers possible. Check that the problem asks for acceleration and the work correctly uses $a = F \div m$.

Suggested use. Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color or with a revision mark.